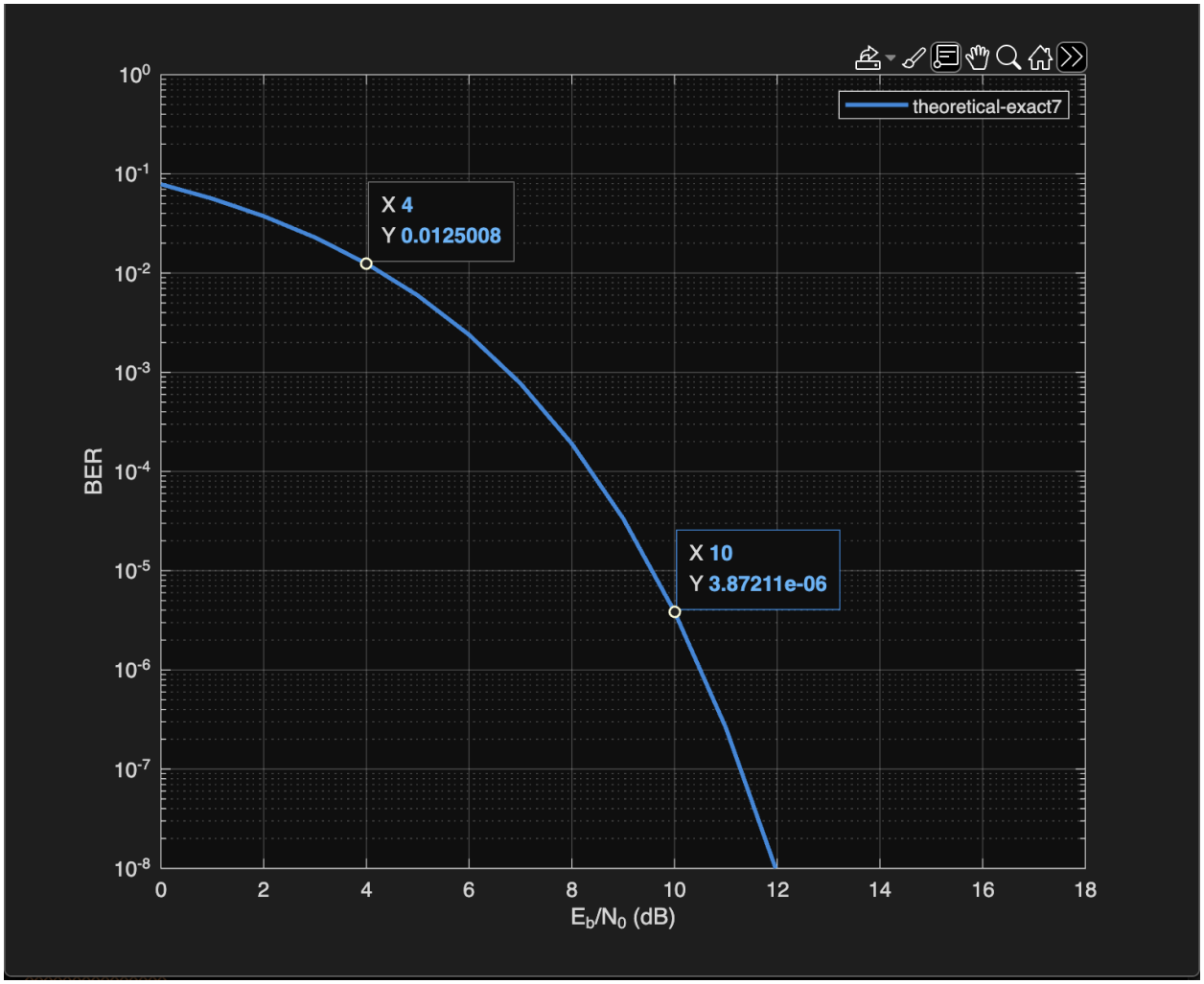
CYSE 425 - Lab 2

Harshith Motipalli
Professor Gebril

Step 8 - E_b/N_0 (dB) = 0:18 dB

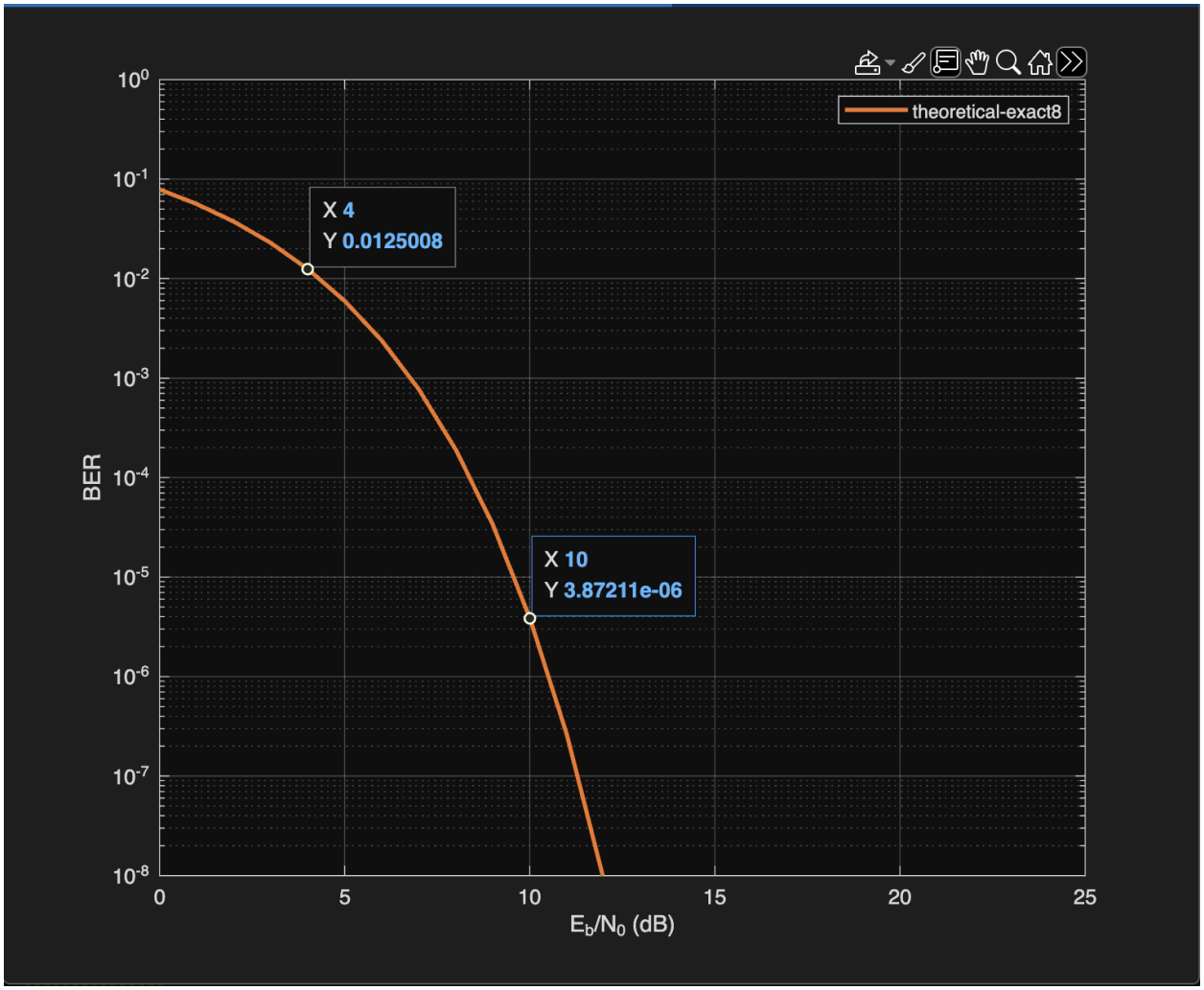


1) **What is the corresponding BER for Eb/N0 = 4 dB**
   The corresponding BER for when E_b/N_0 = 4 DB is 0.0125008

2) **What is the corresponding BER for Eb/N0 = 10 dB**
   The corresponding BER for when E_b/N_0 = 10 DB is $3.87211 * 10^{-6}$

3) **What is your conclusion/observation?**
   The graph shows that the Bit Error Rate decreases as E_b/N_0 increases. At 4 dB, the BER is 0.0125008, while at 10 dB it decreases to approximately $3.87211 * 10^{-6}$. This shows that increasing the signal energy relative to the noise improves the reliability of the communication system because fewer bits are received incorrectly.

Step 9 - E_b/N_0 (dB) = 0:25 dB

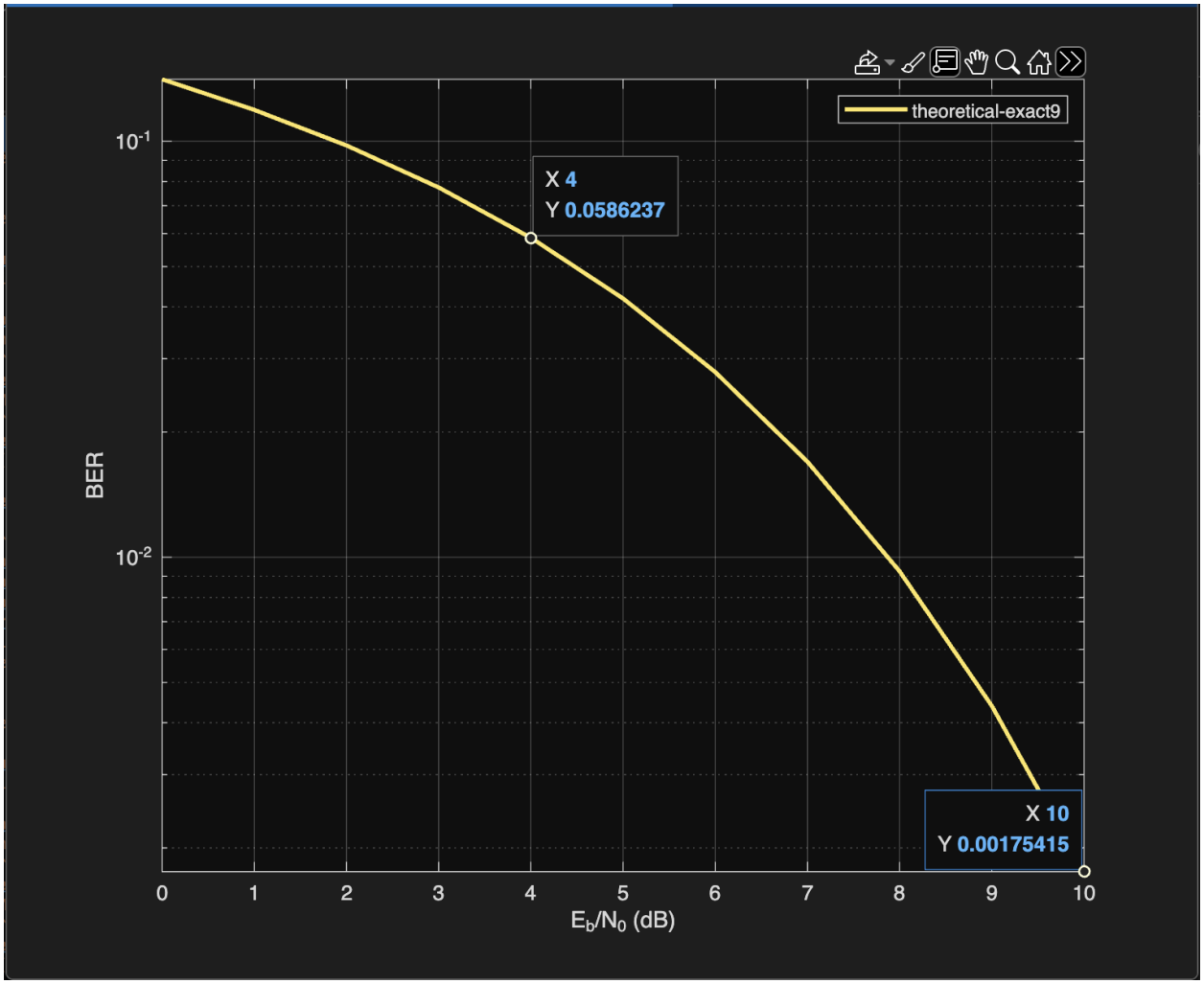


1) **What is the corresponding BER for Eb/N0 = 4 dB**
   The corresponding BER for when E_b/N_0 = 4 DB is 0.0125008

2) **What is the corresponding BER for Eb/N0 = 10 dB**
   The corresponding BER for when E_b/N_0 = 10 DB is $3.87211 * 10^{-6}$

3) **What is your conclusion/observation?**
   Increasing the E_b/N_0 range from 0 - 18 dB to 0 - 25 dB does not change the BER at the same Eb/N0 values because the modulation and channel settings remain unchanged. The larger range simply shows that the BER continues decreasing toward zero as E_b/N_0 increases.

Step 10 - E_b/N_0 (dB) = 0:10 dB | Modulation Type: QAM | Modulation Order: 16

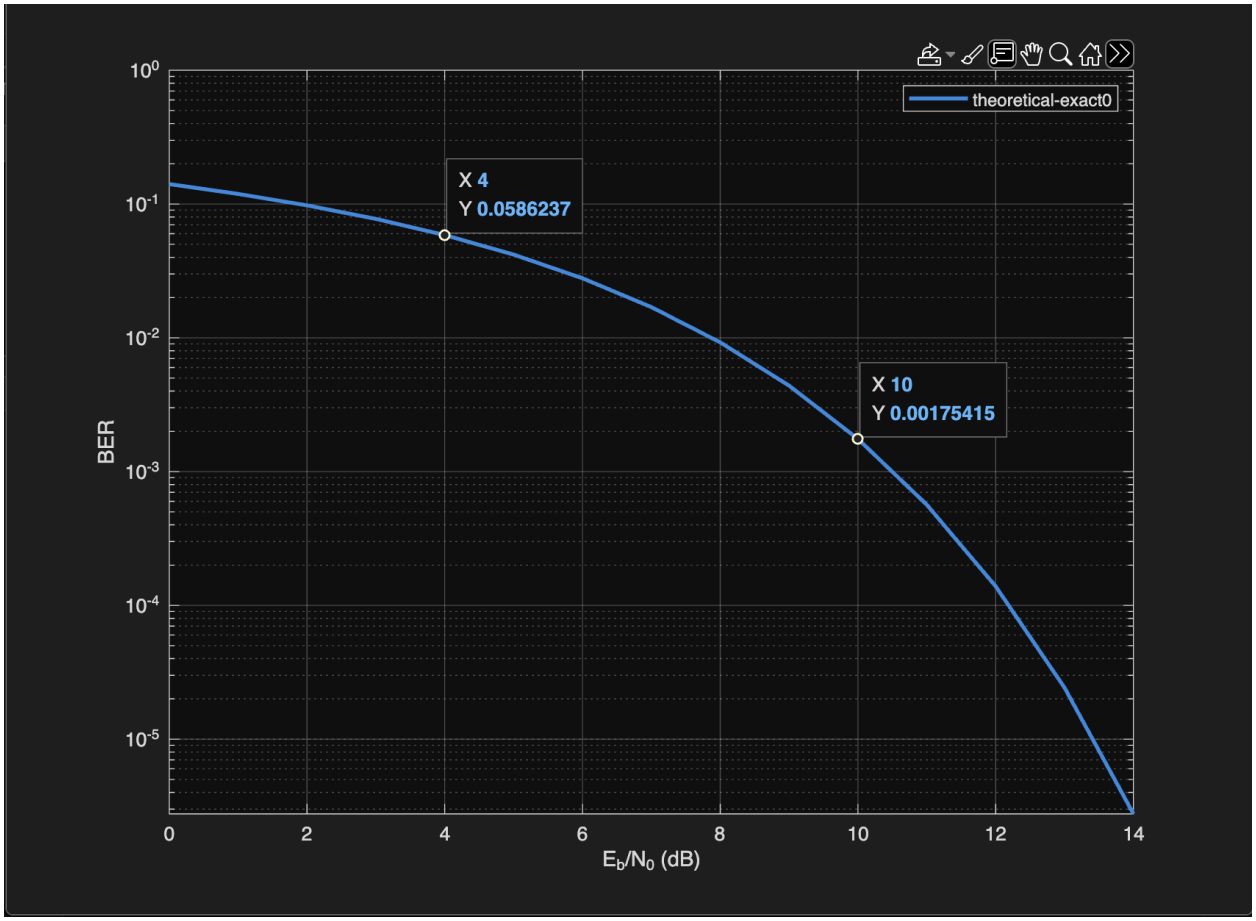


1) **What is the corresponding BER for Eb/N0 = 4 dB**
   The corresponding BER for when E_b/N_0 = 4 DB is 0.0586237

2) **What is the corresponding BER for Eb/N0 = 10 dB**
   The corresponding BER for when E_b/N_0 = 10 DB is 0.00175415

3) **What is your conclusion/observation?**
   For the 16-QAM, the BER decreases as E_b/N_0 increases. At 4 dB, the BER is 0.0586237, while at 10 dB it decreases to 0.00175415. This shows that increasing Eb/N0 reduces the effect of noise and decreases the number of bit errors.

Step 11 - E_b/N_0 (dB) = 0:14 dB | Modulation Type: QAM | Modulation Order: 16

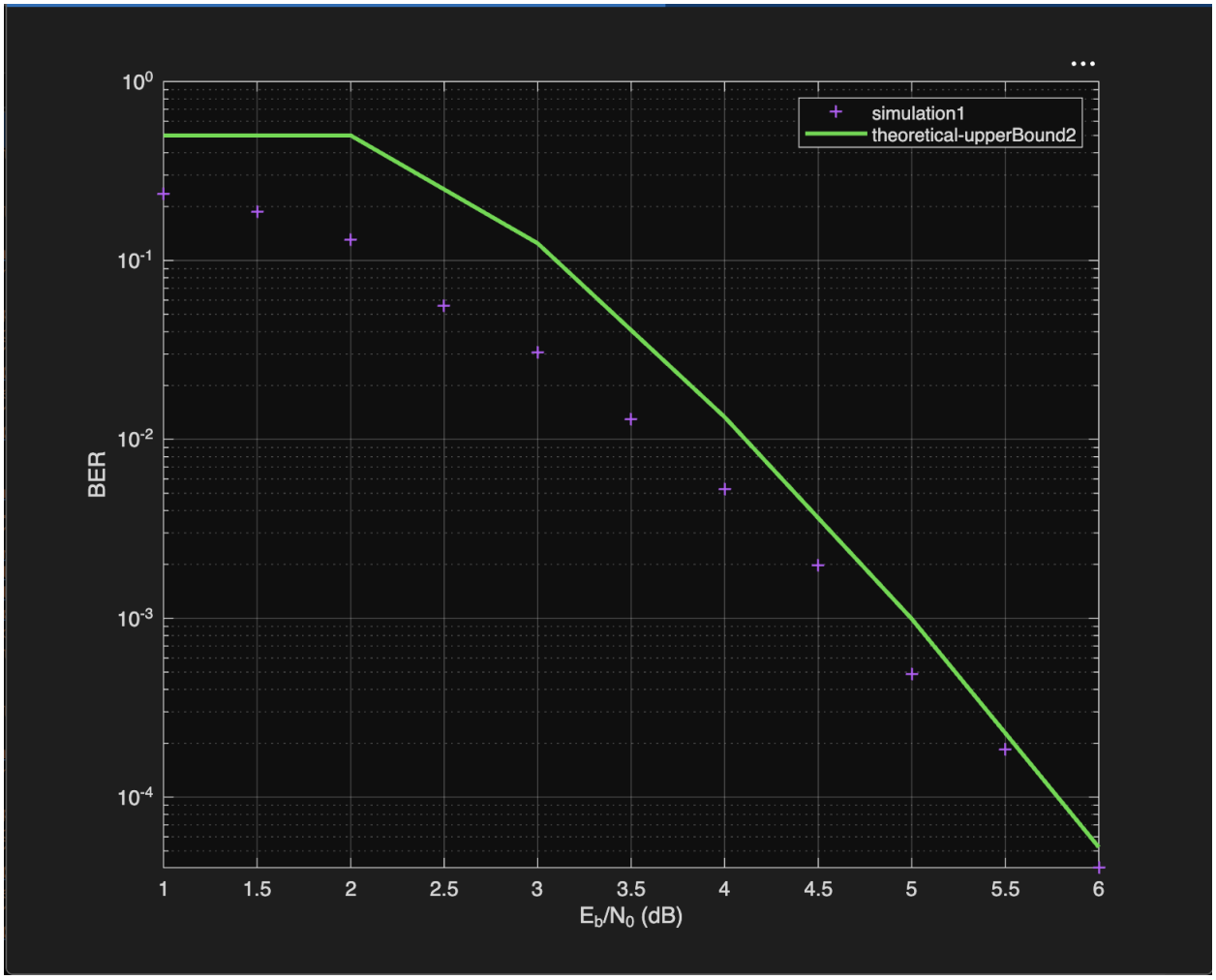


1) **What is the corresponding BER for Eb/N0 = 4 dB**
   The corresponding BER for when E_b/N_0 = 4 DB is 0.0586237

2) **What is the corresponding BER for Eb/N0 = 10 dB**
   The corresponding BER for when E_b/N_0 = 10 DB is 0.00175415

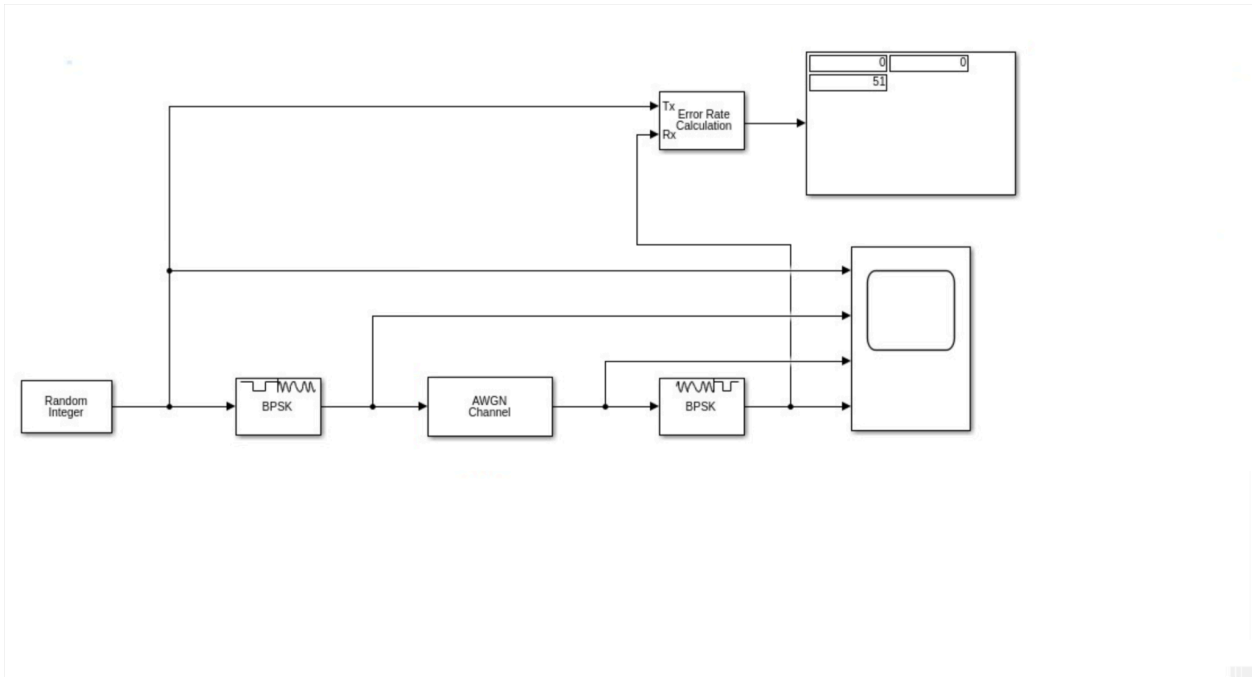
3) **What is your conclusion/observation?**
   Extending the E_b/N_0 range to 14 dB shows that the BER continues to decrease as E_b/N_0 increases. At 4 dB, the BER is 0.0586237, while at 10 dB it decreases to 0.00175415. This shows that increasing the signal energy relative to the noise improves the reliability of the communication system by reducing the number of bit errors.

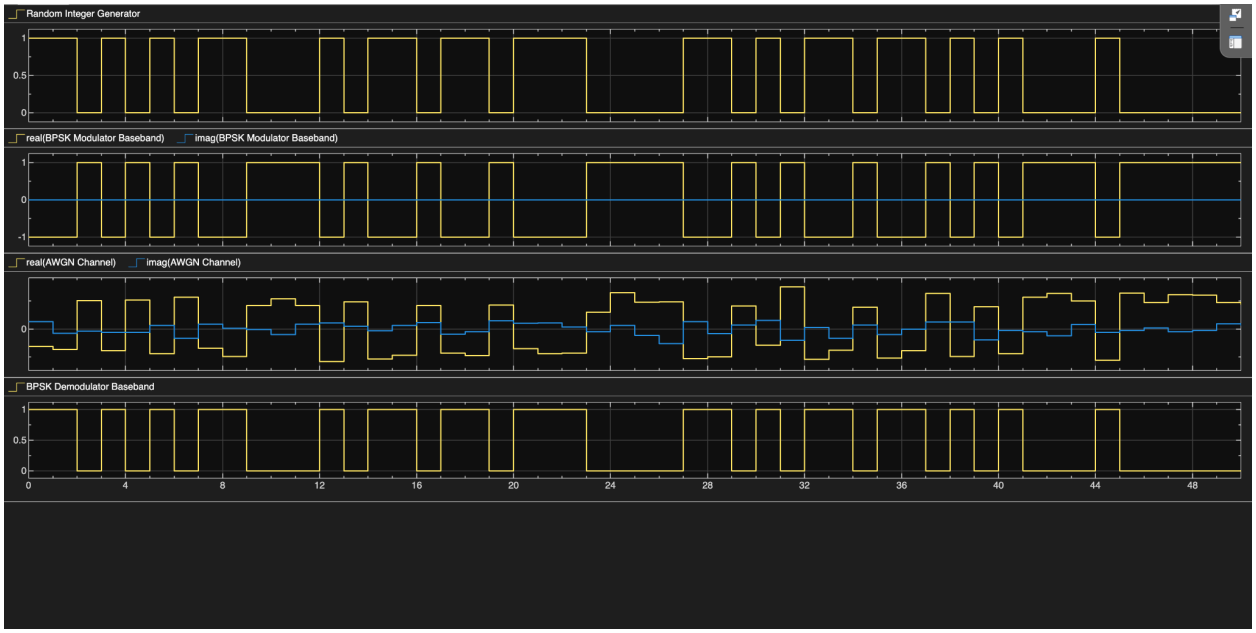
Step 12 - Monte Carlo E_b/N_0 = 1:0.5:6 dB | Theoretical E_b/N_0 = 1:6 dB | Modulation Order: 4



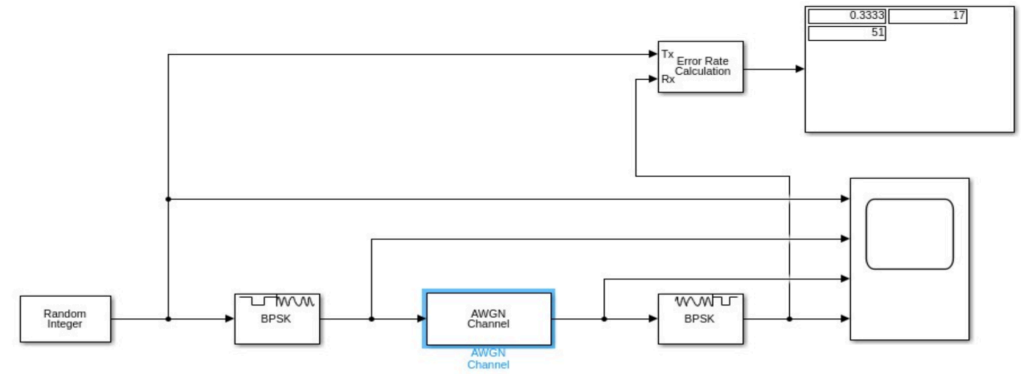
Step 13 - BPSK Modulation & Demodulation**:**

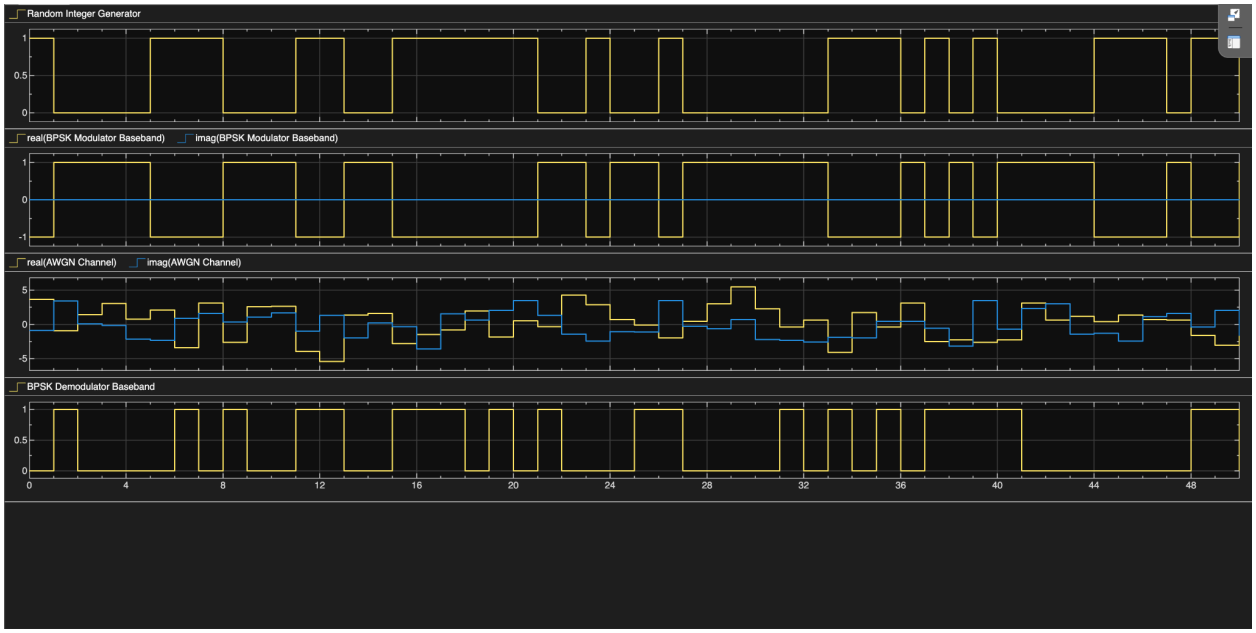
SNR = 10 dB





SNR = -10 dB





1) **Explain the indication of the three numbers 51 0 0 you got on the display for Eb/N0 = 10 dB**
   The three numbers represent the results from the Error Rate Calculation block. A total of 51 bits were compared, 0 bit errors occurred, and the resulting BER was 0. Considering that no bits were received incorrectly, the BER is 0/51 = 0. At 10 dB, the signal is strong enough relative to the noise for the receiver to correctly detect all transmitted bits.

2) **Based on the simulation results, what is the difference for Eb/N0 = -10 dB? Why?**
   At E_b/N_0 = -10 dB, the simulation produces significantly more bit errors than at 10 dB. The display shows approximately 17 errors out of 51 bits, giving a BER of approximately 0.3333. This occurs because the noise is much stronger relative to the signal at -10 dB, making it more difficult for the BPSK demodulator to correctly determine the transmitted bits.